

ED-FEED: An AI-Driven Intelligent Student Feedback Analysis and Insight Generation System

Evaluation of the BERT Model

In this phase of the project, the performance of the sentiment analysis model was evaluated using student feedback data, with a focus on both feature extraction quality and classification accuracy. The dataset was first examined to identify the relevant columns, where the *comments* column was selected as the input text and *sentiment_label* as the target variable. To ensure data quality, missing values in the text column were handled by replacing them with empty strings, and all entries were converted into string format for smooth processing. The text data was then transformed into numerical form using the TF-IDF (Term Frequency-Inverse Document Frequency) vectorizer, which captures the importance of words based on their frequency and relevance. The vectorizer generated approximately 10,000 features, representing key terms from the feedback data. Important domain-specific words such as “hard,” “helpful,” “easy,” and “difficult” were present in the feature set, indicating that the model effectively captures meaningful academic feedback.

For classification, a Logistic Regression model was used and evaluated using standard performance metrics, including precision, recall, F1-score, and accuracy. The model achieved an overall accuracy of approximately 84%, demonstrating strong and reliable performance. It performed particularly well in identifying negative feedback, achieving high recall, which indicates its ability to correctly detect most negative sentiments. The performance for neutral and positive classes was slightly lower, suggesting minor misclassification, especially in cases where the sentiment was mixed or less explicit. The confusion matrix further supported these findings, showing that the majority of predictions were correct, with only a small amount of confusion between neutral and positive classes.

To further evaluate the system’s capability in handling complex language, a transformer-based model, DeBERTa, was tested using zero-shot classification. Unlike traditional machine learning models, this model can understand context, semantic meaning, and subtle variations in language without requiring task-specific training. The results showed that DeBERTa performed better in handling complex sentences, sarcasm, and contextual nuances in the feedback. A comparison between the Logistic Regression model and the DeBERTa model revealed several mismatches in predictions, indicating that the traditional model has limitations in capturing deeper contextual relationships in text.

Overall, the evaluation demonstrates that the Logistic Regression model provides a strong baseline with good accuracy and balanced performance across sentiment classes. However, the comparison with the transformer-based model highlights the importance of contextual understanding in sentiment analysis. This suggests that incorporating advanced models such as transformers can further improve the system’s performance and provide more accurate and meaningful insights from student feedback.

RESULT

Classification Report

Class	Precision	Recall	F1-Score	Support
Negative	0.83	0.90	0.86	11807
Neutral	0.81	0.82	0.81	11807
Positive	0.88	0.79	0.84	11807

Overall Performance

Metric	Value
Accuracy	0.84
Macro Avg F1	0.84
Weighted Avg F1	0.84

The model achieved an overall accuracy of 84%, indicating good performance in correctly classifying student feedback.

Confusion Matrix

Actual \ Predicted	Negative	Neutral	Positive
Negative	10574	917	316
Neutral	1166	9713	928
Positive	1010	1430	9367